Unbounded vertex-position ratios in cubic graphs and local position problems

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Abstract

We prove that the ratio of the upper to lower vertex-position number is unbounded even among connected cubic graphs. A family of order $O(m^3)$ has upper vertex-position number at least m^2 and lower vertex-position number at most $O(m)$; the construction uses staggered connections between rooted geodesic paths. The corresponding ratios for local monophonic position and local mutual visibility are also unbounded. We realize every integer pair (a, b) with $a \geq 2$ and $a \leq b \leq 3a - 1$, and settle the case $a = 1$. An exhaustive computation covers all 556,471 connected cubic graphs through 20 vertices, all connected graphs through nine vertices, and all connected graphs of maximum degree at most three through ten vertices.

1 Definitions and the three questions

Graphs are finite, simple, undirected and connected. Following Thankachy et al. [1, Definition 2.1], an *x-position set* is a set $S \subseteq V(G)$ such that no member of $S \setminus \{v\}$ lies on any shortest x, v-path, for each $v \in S$. Write $p_x(G)$ for its maximum cardinality, and

$$\mathrm{vp}^-(G) = \min_x p_x(G), \qquad \mathrm{vp}(G) = \max_x p_x(G).$$

The observer x cannot belong to an x-position set of size at least two. For nontrivial connected graphs, maximum sets can therefore be taken in $V(G) \setminus \{x\}$; $p_x(K_1) = 1$.

Problem 2.4 of [1] asks whether $\mathrm{vp}(G)/\mathrm{vp}^-(G)$ is bounded over connected graphs. Section 2 supplies the full proof of our original negative answer. James Tuite subsequently asked about bounded degree, local monophonic position and mutual visibility, and realizable pairs $(\mathrm{vp}^-, \mathrm{vp})$ (private communication, 11 October 2026). We answer these questions in Sections 3–5. The bounded-degree question has a sharp qualitative answer: the ratio is bounded for maximum degree at most two and unbounded for maximum degree at most any fixed $\Delta \geq 3$, including the cubic class. The two local variants are unbounded. The pair classification remains partial beyond our stated realization region and finite witnesses.

As Tuite pointed out, trees are already settled in [1, Corollary 3.6]. For a tree T on at least three vertices with ℓ leaves,

$$p_x(T) = \begin{cases} \ell - 1 & x \text{ a leaf}, \\ \ell & x \text{ otherwise}, \end{cases} \qquad \frac{\mathrm{vp}(T)}{\mathrm{vp}^-(T)} = \frac{\ell}{\ell - 1} \leq 2.$$

Thus a new tree enumeration cannot address unboundedness. The exceptions K_1, K_2 both have ratio one. The survey [2, Sections 3.3 and 4.1] places vertex position and monophonic position in their broader setting.

Two elementary facts will be used repeatedly. A sphere about x is an x-position set, since distances strictly increase along an x-geodesic. If $V(G)$ is covered by q geodesics starting at y, then $p_y(G) \le q$: two chosen vertices on the same geodesic violate the definition. Also, an integer function f satisfying $|f(u) - f(v)| \le 1$ on every edge gives $d(u,v) \ge |f(u) - f(v)|$, by summing along a path.

2 The original unboundedness construction

Fix $m \ge 2$. Start with a vertex y and $2m+1$ otherwise disjoint paths from y: H and $P_1, \dots, P_m$ of length $4m$, and Q_j of length $a_j = 2m+2j$, for $1 \le j \le m$. Write $H_t, P_{i,t}, Q_{j,t}$ for their vertices at level t, with all level-zero vertices identified as y. Put $b_j = 2m + j + 1$, and add precisely the edges

$$H_{b_j} Q_{j,b_j}, \qquad Q_{j,a_j} P_{i,a_j} \quad (1 \le i, j \le m).$$

Call the resulting graph G_m, and put $x = H_{4m}$.

Proposition 1. *For $m \ge 2$, the graph G_m has order $7m^2 + 5m + 1$ and satisfies*

$$p_y(G_m) = 2m + 1, \quad \mathrm{vp}^-(G_m) \le 2m + 1, \quad \mathrm{vp}(G_m) \ge m(m-1).$$

Consequently $\mathrm{vp}(G_m)/\mathrm{vp}^-(G_m) \ge m(m-1)/(2m+1)$.

Proof. Every edge changes the level by at most one. Hence each ray is a y-geodesic, and the level of a vertex is its distance from y. The added edges preserve level, so no y-geodesic can use them: a path to level t of length t must increase level at every step. Thus every vertex has a unique y-geodesic, along its own ray. This gives $p_y \le 2m + 1$, with equality by choosing the final vertex of each ray.

Let $M = 2m + 1$. Define $f(y) = M$ and, for positive indices,

$$\begin{aligned} f(H_t) &= \min\{M, 4m - t\}, \\ f(P_{i,t}) &= 2m + (t \bmod 2), \\ f(Q_{j,t}) &= \min\{M, \max\{4m + 1 - t, t - (2j+1)\}\}. \end{aligned}$$

These functions change by at most one on ray edges, including those incident with y. At the first kind of added edge the values are $2m - j - 1$ and $2m - j$; at the second kind they are $2m - 1$ and $2m$. Thus f is 1-Lipschitz on every edge, and $f(x) = 0$.

For $1 \le i \le m$ and $1 \le j < m$, put $z_{i,j} = P_{i,a_j+1}$. Then $f(z_{i,j}) = M$, giving $d(x, z_{i,j}) \ge M$. The path going down H to H_{b_j}, crossing to Q_j, going up to Q_{j,a_j}, crossing to P_i, and taking one more step has length

$$(4m - b_j) + 1 + (a_j - b_j) + 1 + 1 = 2m + 1.$$

The $m(m-1)$ distinct vertices $z_{i,j}$ therefore lie on the sphere of radius M about x. The sphere observation proves the lower bound. The order follows by summing the ray lengths and adding y. □

This proves the negative answer to Problem 2.4. The frozen Lean source formalizes this original family and the two global bounds. The results below are proved here and are not asserted to have been formalized.

3 Maximum degree three and cubic graphs

We replace the common hub by a binary tree and replace each multiway connection at Q_j by connections at successive levels. The target sphere survives because its chosen level compensates for this staggering.

Let $m \geq 2$, put $c = m + 1$, and set

$$L = 2c(m + 2), \quad R = L + 1 - 2c, \quad M = R + c, \quad a_j = 2c(j + 1), \quad b_j = c(j + 2).$$

Let k be the least power of two with $k \geq 2m + 1$. Take a full binary tree of depth $h = \log_2 k$, rooted at y. At $2m+1$ distinct leaves attach rays $H, P_1, \ldots, P_m, Q_1, \ldots, Q_m$, each indexed from $t = 1$ beyond its tree leaf. The H and P_i rays have length L, and Q_j has length $a_j + m$. The other tree leaves have no attached ray. Add the edges

$$H_{b_j} Q_{j,b_j}, \qquad Q_{j,a_j+i-1} P_{i,a_j+i} \quad (1 \leq i, j \leq m).$$

Denote this graph by B_m and put $x = H_L$.

Lemma 2. *The graph B_m is connected, has maximum degree three, and has*

$$N_m = 3m^3 + 13m^2 + 13m + 3 + 2k$$

vertices. It satisfies $p_y(B_m) = k$ and

$$d(x, P_{i,a_j+c}) = M \quad (1 \leq i, j \leq m).$$

In particular $\mathrm{vp}(B_m) \geq m^2$.

Proof. Tree vertices have degree at most three. All added edges have distinct endpoints on each individual ray: the first Q_j exit is at $a_j > b_j$, successive exits are one level apart, and exits on a fixed P_i are $2c$ levels apart. Each such endpoint was an interior degree-two vertex and receives one edge. Ray endpoints remain leaves.

Give tree vertices their depth and ray vertices level $h + t$. Each added edge changes level by zero or one, so every original root-to-leaf path, including its attached ray, remains geodesic. These k paths cover all vertices. They give $p_y \leq k$, and the k leaves of B_m give equality: a leaf cannot be internal to a path ending at a different leaf.

Define $f = M$ on the binary tree, and on the rays define

$$\begin{aligned} f(H_t) &= \min\{M, L - t\}, \\ f(P_{i,t}) &= \min\{M, R + i + \min_{1 \leq q \leq m} |t - (a_q + i)|\}, \\ f(Q_{j,t}) &= \min\{M, \max\{L + 1 - t, t - a_j + R\}\}. \end{aligned}$$

Each ray function is 1-Lipschitz. At its first vertex it equals M: for H and Q_j this follows from $L - M = c - 1 \geq 2$, and for P_i from $a_1 = 4c$. Thus edges to the tree also satisfy the required bound. The identity

$$2b_j = L + 1 + a_j - R$$

shows that the values at H_{b_j}, Q_{j,b_j} are the truncations at M of $L - b_j, L + 1 - b_j$. At the other added edge, the values at Q_{j,a_j+i-1} and P_{i,a_j+i} are respectively $R + i - 1, R + i$. Indeed $a_j \geq 4c$ makes the increasing term in the Q_j maximum dominant there, and $i \leq m < c$ prevents truncation. Hence f is 1-Lipschitz on all edges, and $f(x) = 0$.

At $z_{i,j} = P_{i,a_j+c}$, the nearest point of $\{a_q + i : 1 \leq q \leq m\}$ is $a_j + i$; its distance is $c - i$. This remains true for $j = m$. Therefore $f(z_{i,j}) = M$. For the reverse inequality, follow H down to b_j, cross to Q_j, go up to $a_j + i - 1$, cross to P_i, and go up to $a_j + c$. The length is

$$(L - b_j) + 1 + (a_j + i - 1 - b_j) + 1 + (c - i) = R + c = M.$$

All indices lie on the specified rays: $b_j \le L/2 < L$, $b_j < a_j$, $a_j + i - 1 < a_j + m$, and $a_j + i \le a_j + c \le L - c < L$. The m^2 distinct targets form an x-position set. Finally,

$$|V(B_m)| = (2k-1) + (m+1)L + \sum_{j=1}^{m}(a_j + m) = N_m.$$

□

We now make every degree equal to three. Take two copies of B_m, writing v^0, v^1 for a vertex and its copy. At every degree-two vertex add v^0v^1. At every degree-one vertex u, add four new vertices c_0, c_1, c_2, c_3 in a cycle in that order; join u^0 to c_0, c_1 and u^1 to c_2, c_3. Different leaves receive disjoint cycles. No edges are added at degree-three vertices. Call the resulting graph C_m. The root of B_m has degree two, so y^0y^1 is present.

Theorem 3. *For every $m \ge 2$, C_m is a connected cubic graph of order $2N_m + 4k$ and satisfies*

$$\mathrm{vp}^-(C_m) \le 3k, \qquad \mathrm{vp}(C_m) \ge m^2, \qquad \frac{\mathrm{vp}(C_m)}{\mathrm{vp}^-(C_m)} \ge \frac{m^2}{3k}.$$

The ratio is unbounded among connected cubic graphs. These examples have ratio $\Omega(|V(C_m)|^{1/3})$.

Proof. Every old or new vertex has degree three, and y^0y^1 connects the two copies. There are k degree-one vertices in B_m, giving the asserted order. Collapse the two copies to B_m and each attached four-cycle to its corresponding leaf. Every edge projects to an edge or a stationary step. Consequently distances between vertices in copy zero cannot decrease, and the embedded copy provides equality. The m^2 targets from Lemma 2, in copy zero, are still on the sphere of radius M about x^0.

Write $\lambda(v) = d_{B_m}(y, v)$. In C_m,

$$d(y^0, v^0) = \lambda(v), \qquad d(y^0, v^1) = \lambda(v) + 1.$$

For the second equality, the root edge followed by a copy-one geodesic gives the upper bound. Any path to copy one either uses a copy-switching edge, which projects to a stationary step, or switches via a leaf cycle, whose projected portion is stationary and has at least three edges. Thus its length exceeds its projected length by at least one, giving the lower bound.

For a leaf u with $\lambda(u) = \ell$, its cycle vertices c_0, c_1, c_2, c_3 have distances $\ell+1, \ell+1, \ell+2, \ell+2$ from y^0. The cycle is accessible only through u^0, u^1, so these distances follow from the preceding equalities. For each of the k original root-to-leaf geodesics P, use three y^0-geodesics: its copy-zero version extended through c_0, c_3, its copy-zero version extended through c_1, c_2, and y^0y^1 followed by its copy-one version. These $3k$ geodesics cover $V(C_m)$. Thus $p_{y^0}(C_m) \le 3k$.

As $k < 2(2m+1)$, the displayed ratio exceeds $m^2/(6(2m+1))$, which tends to infinity. Also $|V(C_m)| = O(m^3)$, giving the order-dependent lower growth bound along this family. □

For maximum degree at most two, a nontrivial connected graph is a path or cycle. A path of order at least three has ratio two, and a cycle has $p_x = 2$ at every vertex [1, Corollaries 2.7 and 4.2]. Thus the maximum ratio in that class is two. Cubic graphs belong to every class with maximum degree at most $\Delta \ge 3$, proving unboundedness there. If maximum degree *exactly* $\Delta > 3$ is required, attach $\Delta - 3$ leaves at y^0 in C_m. The old distances are preserved and the root geodesic cover increases by $\Delta - 3$, so the ratio is at least $m^2/(3k + \Delta - 3)$ and is unbounded for each fixed Δ with that convention.

Useful finite-order upper bounds follow from [1, Lemmas 2.5, 2.6, 2.10 and Theorem 2.9]. If $n \ge 2$, minimum degree is δ, maximum degree is Δ, radius is r and diameter is D, then

$$\frac{\mathrm{vp}(G)}{\mathrm{vp}^-(G)} \le \frac{n - r}{\max\{1, \delta, \lceil(\Delta+1)/3\rceil, \lceil(n-1)/D\rceil\}}. \tag{1}$$

For bipartite graphs, $\lceil \Delta/2 \rceil$ can replace $\lceil (\Delta+1)/3 \rceil$; if $r \geq 3$, the numerator can be improved to $n-r-1$ by [1, Theorem 2.11]. In particular the denominator for cubic graphs is at least $\max\{3, \lceil (n-1)/D \rceil\}$. The asymptotic maximum as a function of cubic order remains open: our lower bound is of order $n^{1/3}$ along the displayed family, whereas these general upper bounds can be linear in n.

4 Local monophonic position and local mutual visibility

For these observer-excluding local variants, assume $|V(G)| \geq 2$. We state the localization explicitly, retaining the different path quantifiers in the defining papers. A monophonic path is an induced path. A monophonic position set has no three vertices on any induced path [3, Definition 2.2]; the survey discusses this in Section 4.1 [2]. For a fixed observer x, call $S \subseteq V(G) \setminus \{x\}$ a *local monophonic position set* if, for each $v \in S$, no induced x, v-path contains another member of S. Denote its maximum size by $m_x(G)$, and put $\mathrm{mp}^-_{\mathrm{loc}}(G) = \min_x m_x(G)$ and $\mathrm{mp}^+_{\mathrm{loc}}(G) = \max_x m_x(G)$.

Di Stefano's mutual visibility condition requires the *existence* of a shortest path avoiding the other selected vertices [4, Section 2]. Its local version is: for each $v \in S \subseteq V(G) \setminus \{x\}$, there exists a shortest x, v-path whose internal vertices avoid S. Write $q_x(G)$ for its maximum size, $\mathrm{mv}^-(G) = \min_x q_x(G)$ and $\mathrm{mv}^+(G) = \max_x q_x(G)$. We do not additionally require visibility between two members of S; that would retain the global condition instead of localizing it. These minus signs refer to minima over observers, not to minimum sizes of maximal sets.

Proposition 4. *Both* $\mathrm{mp}^+_{\mathrm{loc}} / \mathrm{mp}^-_{\mathrm{loc}}$ *and* $\mathrm{mv}^+ / \mathrm{mv}^-$ *are unbounded over connected graphs. For the fan* $F_n = K_1 \vee P_n$, $n \geq 2$,

$$\mathrm{mp}^-_{\mathrm{loc}}(F_n) = 2, \qquad \mathrm{mp}^+_{\mathrm{loc}}(F_n) = n.$$

For the original graph G_m,

$$q_y(G_m) = 2m+1, \qquad \mathrm{mv}^+(G_m) \geq m(m-1).$$

Proof. Let w be the universal vertex of F_n, and label its path $v_1, \ldots, v_n$. Every induced w, v_i-path is the single edge, so $m_w = n$. At a path vertex v_t, at most one selected path vertex can lie strictly on each side of v_t: the intervening path segment is induced, and two on the same side violate the local condition. The only additional possible selected vertex is w. Thus $m_{v_t} \leq 3$, and $m_{v_1}, m_{v_n} \leq 2$. The neighborhood of any observer is always a local monophonic position set, since an induced path between adjacent endpoints must be their edge. Hence endpoints have value two and internal path vertices have value three. Together with $m_w = n$, this proves the asserted extrema, including $n = 2$.

In G_m, every y-geodesic is the unique ray path, as proved in Proposition 1. Consequently two selected vertices on one ray cannot be locally visible from y, and the ray endpoints give $q_y = 2m+1$. The $m(m-1)$ targets on the sphere about x are locally visible from x: every shortest path to a target has all internal vertices at smaller distance, so it avoids all other targets. The resulting lower ratio bound is again $m(m-1)/(2m+1)$. □

If one counts the observer in local mutual visibility, it can be adjoined freely, adding one to every q_x. The same family then gives the lower ratio bound $(m(m-1)+1)/(2m+2)$, so the unboundedness conclusion is independent of this counting convention.

5 Realizable pairs

Necessarily $1 \leq a \leq b$ when $a = \mathrm{vp}^-(G)$ and $b = \mathrm{vp}(G)$. By [1, Corollary 2.7], $a = 1$ holds precisely for paths. Consequently the pairs with $a = 1$ are exactly $(1,1)$ and $(1,2)$, represented by K_1 or K_2, and P_3, respectively. Every diagonal pair (a, a) is realized by K_{a+1}.

Thankachy et al. [1, Lemma 2.3] also give the exact pairs $(a, 6a-4)$ for every $a \geq 3$. The following construction realizes a full interval of values for each $a \geq 2$.

Theorem 5. *Every pair of integers $a \geq 2$, $a \leq b \leq 3a-1$ is realized by a connected graph with $\mathrm{vp}^- = a$ and $\mathrm{vp} = b$.*

Proof. It remains to handle $a+1 \leq b \leq 3a-1$. Choose positive integers $r, t \leq a$ with $r+t = b+1-a$. Take a vertex y, independent sets A, D of sizes r, t, and a clique B of size a. Put in all edges between y and A, between A and B, and between B and D, and no other edges. The graph has $b+2$ vertices.

From y, the distance layers are A, B, D; every vertex in an earlier layer lies on a geodesic to every vertex in a later layer. Thus $p_y = \max\{r, a, t\} = a$. From a vertex of A, the layers are $\{y\} \cup B$ and $(A \setminus \{u\}) \cup D$. A selected vertex of B forbids the entire second layer; y forbids $A \setminus \{u\}$ but is incomparable with D. Hence

$$p_u = \max\{a+1, r+t-1, t+1\} \quad (u \in A).$$

From $u \in D$, the layers are B, $A \cup (D \setminus \{u\})$, and $\{y\}$. Selecting a vertex of B forbids both later layers; in their union, y forbids A and is incomparable with $D \setminus \{u\}$. Therefore

$$p_u = \max\{a, r+t-1, t\} = \max\{a, r+t-1\} \quad (u \in D).$$

Finally, from $u \in B$, all other vertices except y are neighbors. The second-layer vertex y is comparable precisely with A. Thus

$$p_u = \max\{r+a-1+t, a+t\} = b \quad (u \in B).$$

The formulas exhibit all extremal choices and exclude larger ones. All four displayed values lie between a and b, with a attained at y and b attained in B. □

The region in Theorem 5 is not a necessary condition. An explicit nine-vertex graph has $(\mathrm{vp}^-, \mathrm{vp}) = (2, 6)$. On $V = \{0, \ldots, 8\}$ take the edge set

$$\{05, 06, 07, 15, 17, 26, 28, 37, 38, 48, 57, 68\}.$$

Here uv means the edge $\{u, v\}$. The exact values in vertex order are

$$(p_0, \ldots, p_8) = (4, 2, 3, 6, 3, 3, 5, 4, 4).$$

Exhaustive computation through order nine proves that nine is the minimum order for this pair. The minimum orders for $(2, b)$, $b = 2, 3, 4, 5, 6$, are respectively $3, 4, 5, 7, 9$. Connected cubic graphs additionally realize $(3, 9)$ at order 18. The complete classification beyond the proved region remains open; in particular, these computations do not decide whether $(2, b)$ is realizable for any $b \geq 7$.

6 Exact finite computations and remaining bounds

Handle K_1 separately with $p_x(K_1) = 1$. For $n \geq 2$ and fixed x, define the geodesic order on $V(G) \setminus \{x\}$ by

$$u \preceq_x v \quad \Longleftrightarrow \quad d(x, u) + d(u, v) = d(x, v).$$

This is a partial order: concatenating geodesic equalities and using the triangle inequality proves transitivity, and a strict relation increases distance from x. Its antichains are exactly the x-position sets that exclude x. Dilworth's theorem [5] therefore computes p_x as $n-1$ minus a maximum matching in the bipartite graph of *all strict comparable pairs*. Using all comparable

pairs, rather than just graph edges, is essential to this formula. All distances and matching decisions use integers. Polynomial-time computability of these position parameters was already established in [1, Theorem 5.7 and Corollary 5.8]. Indeed, matching edges concatenate into disjoint directed chains, since each vertex has at most one predecessor and successor and strict order forbids cycles; a matching of size s gives $n-1-s$ chains. Conversely, linking successive vertices in a chain cover gives a matching of size $n-1$ minus the number of chains.

For completeness, a maximum matching also gives an antichain of that size. Put $N = n-1$ and let the matching size be s. Start at every unmatched left vertex and traverse unmatched edges left to right and matched edges right to left. Write Z_L, Z_R for the reachable vertices in the two copies. No unmatched right vertex is reachable, since that would give an augmenting path. Every reachable matched left vertex is paired with a reachable right vertex, so $|Z_L|-|Z_R| = N-s$. Identifying the two copies with their vertex labels gives $Z_R \subseteq Z_L$: if v_R is reached from u_L with $u \prec_x v$, either v_L is unmatched and is a starting vertex, or it is matched to some w_R. In the latter case $u \prec_x v \prec_x w$ implies the unmatched edge $u_L w_R$, followed by the matched edge $w_R v_L$. Consequently $A = \{v : v_L \in Z_L,\ v_R \notin Z_R\}$ has size $N-s$. If $u \prec_x v$ and u_L is reachable, then v_R is reachable: their edge is either traversable left to right or is the matched edge by which u_L was reached. Thus A is an antichain. Together with the chain cover, this proves the matching formula directly.

We generated one representative of each connected graph in the specified classes with `geng` from nauty 2.9.1 [6]. Connectivity and degree constraints were imposed at generation. Table 1 gives every connected cubic order through 20, with no girth, bipartiteness or connectivity-strength restriction.

Order	4	6	8	10	12	14	16	18	20
Graphs	1	2	5	19	85	509	4,060	41,301	510,489
Maximum ratio	1	1	4/3	2	2	7/3	8/3	3	3

Table 1: Exact extrema over all connected cubic graphs at each order.

There are 273,193 connected graphs of orders one through nine in the second search, and 2,571 connected graphs of maximum degree at most three of orders one through ten in the third. Their exact maximum ratios by order are, respectively,

$$(1,1,2,2,2,2,5/2,5/2,3), \qquad (1,1,2,2,2,2,2,5/2,5/2,5/2).$$

The data record every p_x, every pair frequency and explicit graph witnesses. A fresh enumeration reproduces every graph6 input. An independent exact antichain search recomputes all 832,235 graph records and all 13,496,353 observer values, without using matching. The tree formula holds on all 81 subcubic trees of orders three through ten. Exhaustive subsets, induced-path enumeration, and reachability in the shortest-path directed acyclic graph check the two local variants on all 142 nontrivial connected graphs through order six and fans with $2 \le n \le 10$.

The construction checks verify every edge of the Lipschitz functions, every geodesic in the stated covers, and every selected sphere vertex for $2 \le m \le 12$, with additional cubic checks at $m = 20, 40$. For example, C_{40} has 427,670 vertices, a sphere containing 1,600 selected vertices, and a cover by 384 root geodesics; its ratio is therefore at least 25/6. These are certified lower and upper bounds for the two parameters, not computed equalities for those large graphs. The realization formulas are checked exactly for every $1 \le r, t \le a$ with $2 \le a \le 15$, together with the diagonal examples for $1 \le a \le 30$.

The proofs settle boundedness for each fixed maximum-degree class and both local variants. The quantitative cubic growth rate and the full pair classification are the remaining questions. In particular, a finite maximum of three through cubic order 20 does not imply an all-order upper bound of three.

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